



LOGIN CONTROL

Understanding Windows & Active Directory Authentication



1 USER TYPES AT LOGIN

Windows supports different types of accounts that can be used to sign in. The login format you use determines where the account is validated and what access you receive.



LOCAL USER

`.\Ahmed`

Authenticates against the local computer's SAM (Security Account Manager) database.



Used for standalone systems or when not joined to a domain.



DOMAIN USER

`NETPOINT\Ahmed`

Authenticates against the Active Directory domain controller.



Used for everyday access to domain resources.



DOMAIN ADMINISTRATOR

`NETPOINT\Administrator`

Authenticates as a member of the Domain Admins group.



Used for administration and managing domain resources.

2 DOMAIN USER ≠ DOMAIN ADMIN

Both authenticate to the domain, but their group membership determines their permissions. Domain Users have limited rights by design, while Domain Admins have full control.

DOMAIN USERS GROUP



Limited Rights

- Standard access to resources
- Cannot install software
- Cannot change security settings
- Cannot manage users or groups



DOMAIN ADMINS GROUP



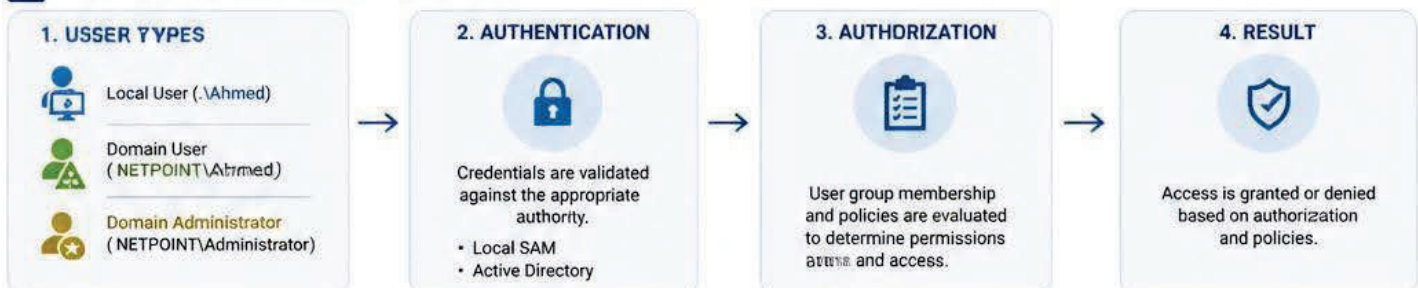
Full Rights

- Full control of domain
- Can install and configure systems
- Can manage users and groups
- Can change security settings

3 QUICK REFERENCE TABLE

USER TYPE	LOGIN FORMAT EXAMPLE	AUTHENTICATION SCOPE	DEFAULT RIGHTS	TYPICAL USE CASE
 Local User	<code>.\Ahmed</code>	 Local computer (SAM database)	Limited to local machine administration (if in local Administrators group)	Standalone PCs, kiosks, test machines
 Domain User	<code>NETPOINT\Ahmed</code>	 Active Directory domain controller	Standard user rights (least privilege)	Daily work, accessing domain resources
 Domain Administrator	<code>NETPOINT\Administrator</code>	 Active Directory domain controller	Full administrative rights across the domain	Administration, management tasks

4 LOGIN ARCHITECTURE FLOWCHART



KEY TAKEAWAYS

- ✓ The login format determines where authentication occurs.
- ✓ Domain Users have limited rights by default.
- ✓ Domain Admins have full control—use with caution.
- ✓ Always follow the principle of least privilege.
- ✓ Use separate admin accounts for administration tasks.

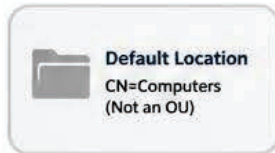
OBJECT REDIRECTION

Automating Computer and User Account Placement with REDIRCMP & REDIRUSR

1 DEFAULT BEHAVIOR VS REDIRECTION

Default Behavior

When a new computer or user joins the domain, the account is created in the default domain container.



After Redirection

With redirection configured, new accounts are automatically created in the designated Organizational Unit.



2 WHY REDIRECT OBJECTS?



Automatic GPO Application

Objects are created in OUs where the correct Group Policy Objects (GPOs) are already linked.



Organized Structure

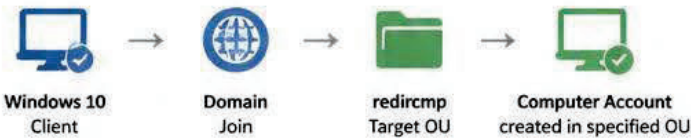
Keeps the Active Directory (AD) clean, logical, and easier to navigate.



Scalable Management

Ideal for environments with many users and computers joining the domain.

3 REDIRCMP FOR COMPUTERS



Command Syntax:

```
redircmp "OU=Computers,OU=LAB,DC=netpoint,DC=local"
```

Run the above command on the client BEFORE joining the domain.

4 REDIRUSR FOR USERS

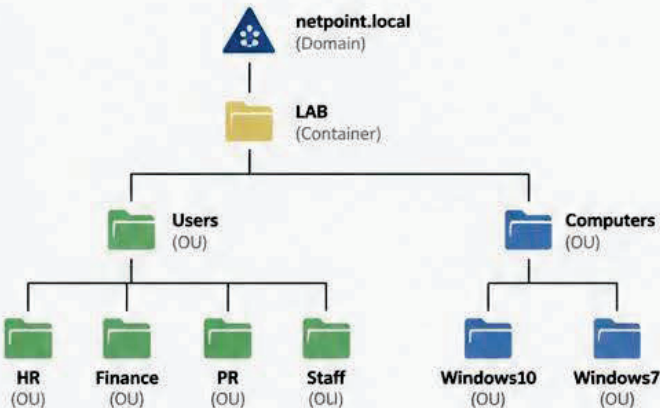


Command Syntax:

```
redirusr "OU=Users,OU=LAB,DC=netpoint,DC=local"
```

Run the above command on the client BEFORE first user logon to the domain.

5 EXAMPLE OU STRUCTURE



6 REDIRCMP ≠ MOVE

Aspect	Manual Move	Default Redirection
When it Happens	After account is created in CN=Computers	During account creation
Admin Effort	Manual and repeated	Automatic and seamless
Risk	GPOs may not apply immediately	Correct GPOs apply from the start
Audit/History	Shows move operation in AD logs	Shows account created in target OU
Best For	Occasional or one-time adjustments	Standard, long-term management

KEY TAKEAWAYS



- By default, objects are created in the CN=Computers container (not an OU).
- Use `redircmp` and `redirusr` to ensure objects are created in the right OU from the start.
- This ensures automatic GPO application, better organization, and easier administration.
- Plan your OU structure first, then implement redirection.

COMMAND QUICK REFERENCE



Syntax:

```
redircmp "OU=Computers,OU=LAB,DC=netpoint,DC=local"
```

Use before domain join on the computer.



Syntax:

```
redirusr "OU=Users,OU=LAB,DC=netpoint,DC=local"
```

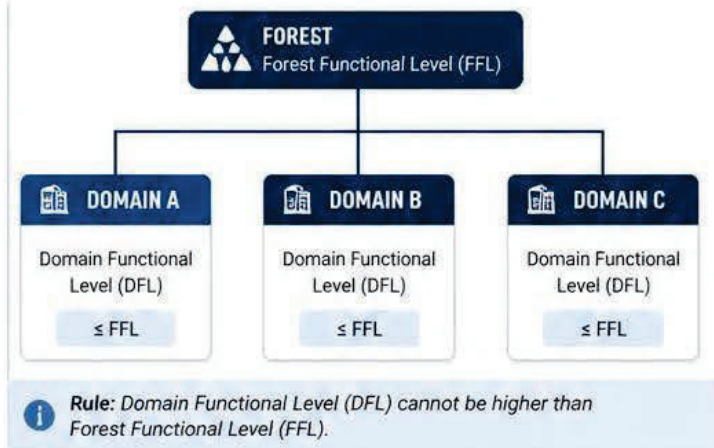
Use before first logon on the user's client.

FUNCTIONAL LEVELS & RECYCLE BIN

Managing Active Directory Features and Object Recovery

1 FOREST FUNCTIONAL LEVEL VS DOMAIN FUNCTIONAL LEVEL

The Forest Functional Level (FFL) is set at the forest root and applies to the entire forest. Each domain has its own Domain Functional Level (DFL), which cannot be higher than the FFL.



2 WHY RAISE FUNCTIONAL LEVELS

Raising functional levels enables newer features and improvements.



Access to New Features

Enables features like AD Recycle Bin, Fine-Grained Password Policies, and Privileged Access Management.



Improved Security

Leverages modern authentication, encryption, and security enhancements.



Performance & Scalability

Benefits from directory optimizations and larger environment support.



Better Interoperability

Ensures compatibility with modern Windows Server and Microsoft services.



Support & Future Readiness

Meets Microsoft support requirements for current and future products.

3 SUPPORTED FUNCTIONAL LEVELS TABLE

Windows Server Version	Domain Functional Level (DFL)	Forest Functional Level (FFL)
Windows Server 2008 R2	Windows Server 2008 R2	Windows Server 2008 R2
Windows Server 2012	Windows Server 2012	Windows Server 2012
Windows Server 2016	Windows Server 2016	Windows Server 2016
Windows Server 2019	Windows Server 2019	Windows Server 2019
Windows Server 2022	Windows Server 2022	Windows Server 2022

i You can raise domain functional level independently. Forest functional level can be raised only after all domains are at the desired level.

4 ACTIVE DIRECTORY RECYCLE BIN

Recycle Bin allows you to recover deleted AD objects.



i Requires: Forest Functional Level ≥ Windows Server 2008 R2

5 HOW TO RESTORE

You can restore deleted objects using PowerShell or AD Administrative Center.

A. POWERSHELL

List deleted objects in the Recycle Bin:

```
Get-ADObject -Filter 'isDeleted -eq $true' -IncludeDeletedObjects
```

Restore an object:

```
Restore-ADObject -Identity "CN=John Doe,OU=Users,DC=contoso,DC=com"
```

Optional: Restore to a specific location:

```
Restore-ADObject -Identity "CN=John Doe,OU=Users,DC=contoso,DC=com" -TargetPath "OU=Users,DC=contoso,DC=com"
```

B. AD ADMINISTRATIVE CENTER

- 1 Open AD Administrative Center.
- 2 Click Deleted Objects.
- 3 Locate the object you want to restore.
- 4 Right-click the object and click Restore. Choose the target location if needed.

KEY TAKEAWAYS

- ✓ FFL applies to the entire forest; DFL applies to each domain.
- ✓ You can raise DFL independently, but FFL only after all domains are ready.
- ✓ Higher functional levels unlock new features and better security.
- ✓ AD Recycle Bin helps recover accidentally deleted objects.
- ✓ Always test and plan before raising functional levels in production.

IMPORTANT NOTES

- Raising functional levels is one-way and cannot be undone.
- Ensure all domain controllers are compatible and healthy.
- Raise DFL first, then FFL.
- Enable AD Recycle Bin only after raising FFL to Windows Server 2008 R2 or higher.
- Backup before making any functional level changes.

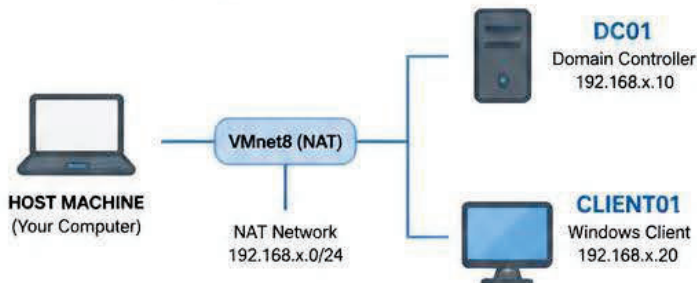


OFFLINE DOMAIN JOIN (ODJ)

Domain Joining Without Live DC Connectivity

1 LAB NETWORK OVERVIEW

VMware NAT Topology



All VMs are on the same LAN (VMnet8 – NAT) and can communicate when network connectivity is enabled.

2 IP/DNS CONFIGURATION TABLE

Setting	DC01 (Domain Controller)	CLIENT01 (Windows Client)
IP Address	192.168.x.10	192.168.x.20
Subnet Mask	255.255.255.0	255.255.255.0
Default Gateway	192.168.x.2 (VMnet8 NAT)	192.168.x.2 (VMnet8 NAT)
Preferred DNS	127.0.0.1 (or 192.168.x.10)	192.168.x.10 (DC01)
Alternate DNS	-	-

Why does the client use the DC as DNS?
The client uses the Domain Controller (DC01) as its DNS server so it can resolve domain names (e.g., DC01, CLIENT01, domain.local) using the AD-integrated DNS hosted on the DC.

3 WHAT IS OFFLINE DOMAIN JOIN?

Offline Domain Join (ODJ) allows a client to join a domain without needing network connectivity to a Domain Controller at the time of join.



5 DOMAIN MEMBERSHIP ≠ DC CONNECTIVITY



A computer can be a member of the domain without having current network connectivity to the Domain Controller.

6 CAN YOU LOG IN?

Cached Credentials (Previously Logged In)
YES – You can log in using accounts that have logged in before. Windows caches domain credentials locally for offline logon.

First-Time Login (Never Logged In)
NO – You cannot log in with a domain account that has never logged in on this machine. The computer cannot validate new credentials without contacting a Domain Controller.

Once the client connects to the DC, normal authentication resumes for all users.

4 STEP-BY-STEP LAB PROCESS

1 Create ODJ file on DC01

```
djoin /provision /domain domain.local /machine CLIENT01  
/savefile C:\ODJ\CLIENT01.odj /reuse
```

2 Disconnect DC01 network (simulate offline scenario)

- Disable Network Adapter
or remove network cable

3 Apply offline join on CLIENT01

```
djoin /requestODJ /loadfile C:\ODJ\CLIENT01.odj /windowspath %SystemRoot%\localos
```

4 Restart CLIENT01

```
shutdown /r /t 0
```

5 Reconnect DC01 network

- Enable Network Adapter
or reconnect network cable

6 Verify connectivity from CLIENT01

```
ping DC01.domain.local  
nltest /dsgetdc:domain.local
```

★ KEY TAKEAWAYS

- ✓ ODJ allows domain joining without live DC connectivity.
- ✓ The provisioning file (.odj) contains encrypted data used for the join.
- ✓ Domain membership and DC connectivity are NOT the same.
- ✓ Cached credentials allow logon for previously used accounts.
- ✓ Reconnecting to the DC restores full functionality and policy application.

⚠ IMPORTANT NOTES

- Use ODJ only when absolutely necessary (e.g., branch offices, sealed networks, or imaging scenarios).
- Keep the .odj file secure and use it only once if possible.
- Ensure time synchronization between client and DC for successful join.
- After reconnection, allow time for Group Policy and updates to apply.





DOMAIN JOIN PERMISSIONS

Controlling Who Can Join Computers to the Domain



1 DEFAULT DOMAIN JOIN PERMISSIONS



By default, members of the **Authenticated Users** group can join up to **10** computers to the domain.



This default setting exists for backward compatibility and convenience but can pose a security risk if not properly managed.

2 WHO CAN JOIN COMPUTERS?

DEFAULT (Authenticated Users)	RESTRICTED (Specific Groups Only)	DELEGATED (Department Admins)
- Any authenticated user can join computers - Up to 10 computers per user (by default) - Least secure option	- Only members of specific security groups can join computers - More secure - Recommended approach	- Join rights delegated to department or OU admins - Limit scope to specific OUs or departments - Scalable and manageable

3 ms-DS-MachineAccountQuota

The **ms-DS-MachineAccountQuota** attribute defines the maximum number of computer accounts a standard user can create (i.e., join) in the domain.



Default Value

The default quota is **10** computers per user.



Check the Current Value

Use the following PowerShell command:

```
Get-ADDomain | Select-Object ms-DS-MachineAccountQuota
```



Modify the Value

Use the following PowerShell command:

```
Set-ADDomain -Identity "yourdomain.com" `
-Replace @(ms-DS-MachineAccountQuota=0)
```

- Set to **0** to prevent non-admin users from joining any computers.

4 RESTRICTING DOMAIN JOIN PERMISSIONS

To restrict who can join computers to the domain, follow these steps.



USING ADUC (Active Directory Users and Computers)

- Create a Security Group**
Create a global security group (e.g., "Domain Join Admins").
- Remove "Authenticated Users" from Domain Join Permissions**
 - Open ADUC.
 - Right-click the domain root and select Properties.
 - Go to the Security tab → Advanced.
- Add Specific Group to Domain Join Permissions**
 - Click Add.
 - Add your security group (e.g., Domain Join Admins).
 - Allow the following permissions: **Create Computer Objects**.



USING POWERSHELL

- Create a Security Group**
`New-ADGroup -Name "Domain Join Admins" -GroupScope Global -GroupCategory Security`
- Remove Authenticated Users**
`dsacl "DC=yourdomain,DC=com" /G "Authenticated Users":CR;computer /R`
- Add Your Security Group**
`dsacl "DC=yourdomain,DC=com" /G "Domain Join Admins":CR;computer /R`



Note: The permission "Create Computer Objects" (CR) on the domain root allows the group to join computers to the domain.

5 DELEGATING DOMAIN JOIN RIGHTS



You can delegate domain join permissions to department admins or to specific OUs.

- Create a security group for each department (e.g., "HR Join Admins").
- Delegate "Create Computer Objects" permission on the domain root or specific OU where computers will reside.
- Add department admins to the corresponding group.



Scope It Correctly

Delegate permissions only where computers will be created. Avoid granting rights at the domain root if OU-level delegation is sufficient.

6 BEST PRACTICES



Restrict domain join to authorized personnel only.



Monitor domain join activity via security logs (Event ID 4741 – A computer account was created).



Use security groups to manage permissions for scalability.



Document your domain join policy and delegated permissions.



Test changes in a lab environment before applying to production.



KEY TAKEAWAYS

- By default, **Authenticated Users** can join up to **10** computers.
- The **ms-DS-MachineAccountQuota** controls how many.
- Restrict and delegate domain join rights to reduce risk.
- Use groups and least-privilege principles for management.



COMMON MISTAKES

- Leaving default permissions (**Authenticated Users**) in place.
- Setting **ms-DS-MachineAccountQuota** too high or unlimited.
- Granting join rights at the domain root unnecessarily.
- Not monitoring or auditing domain join activity.

